

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)	(i)		16.8%/ 17%		1		1		
		(ii)		UK is close(r) to {Europe / other countries}/ Hawaii great(er) distance from North America / species can easily {migrate/ fly} to UK / Hawaii isolated/ Hawaii relatively young		1		1		
	(b)			Eukarya / Eukaryota/ Eukaryote (1) Order and Family (1) <i>Himatione</i> and <i>sanguinea</i> (1) correct spelling and cases should be used	2	1		3		
	(c)	(i)		(apapane) finch (1) oriole + tanager Both correct = 1 mark			2	2		
		(ii)		Any three (×1) from: In the context of the two most closely related species A. The {base/ DNA} <u>sequences</u> are {most/ more} similar/ {most/ more} bases which are complementary (1) B. {most/ more} hydrogen bonds will form (1) C. The lower difference in separation temperature/ highest separation temperature (1) D. They share the {most/ more} recent common ancestor (1) Cannot award D in isolation Accept reverse answer for least closely related		1	2	3		

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	(d)	(i)		As they are unable to {interbreed/ reproduce with each other} to produce fertile offspring. (1)	1			1		
		(ii)		Any four (×1) from: A. Adaptive radiation/ divergent evolution (1) B. Variation in beak shape evolved from the original species/ variation in beak shape in the population of the original species (1) C. As the population (of the ancestral species) increased there would have been increased competition for food/ owtte (1) D. Ref to particular beak shape for particular food / variety of beaks for variety of foods (1) E. would have a selective advantage/ Natural selection/ survival of fittest/ or description of (1)	1	1	2	4		
				Question 4 total	4	5	6	15	0	0